

Custom Exam Paper (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)(i)	0.2 or $\frac{1}{5}$	1	B1
1(a)(ii)	$\frac{3}{2}$ or $1\frac{1}{2}$ or 1.5	1	B1
1(b)	$\frac{2}{x^2}$ or $2x^{-\frac{9}{2}}$	2	B2 for correct answer
			M1 for 2 or $x^{-\frac{9}{2}}$ seen
2(a)	43945	2	M1 for $75000 \times (1 - 0.125)^4$ or 75000×0.875^4
2(b)	41.4%	2	M1 for $\frac{75000 - 43945.31}{75000} \times 100$ or $1 - 0.875^4$
3(a)	$10\sqrt{3}$	2	M1 for $12\sqrt{3}$ or $2\sqrt{3}$
3(b)	$7 - 3\sqrt{3}$	3	M1 for multiplying numerator and denominator by $2 - \sqrt{3}$
			M1 for numerator $10 - 5\sqrt{3} + 2\sqrt{3} - 3$